

+20 12 7247 3302  
Cairo, Egypt  
hamzahassanain067@gmail.com

# Hamza Mohammed Hassanain

Software Engineer

github.com/HamzaHassanain  
gitlab.com/HamzaHassanain  
linkedin.com/in/hamza-hassanain-ha067  
dev.to/hamzahassanain0

## EDUCATION

B.Sc. Computer and Information Sciences, Suez Canal University  
AWS Certified Solutions Architect – Associate, Amazon Web Services

2022 – 2026  
[credly.com/verify](https://www.credly.com/verify)

## EXPERIENCE

Founding Software Engineer Nov 2025 – Present  
Repovive (YC F’26) Full Time – United States, Remote  
Where experts solve problems AI can’t: rated live contests, 140+ IOI-grade problems, and 10,000+ users.

- Joined as the **first engineer** and built the core systems from the early days through YC.
- Led design of the distributed judge (judge1 on **Judge0**); scaled submissions **10x (50 to 500+ concurrent)**.
- Cut infrastructure cost **95% (\$600 to \$25/mo)** by migrating resource-heavy tasks from serverless to Docker/Nginx/Redis.
- Isolated **20+ languages** in resource-limited Docker images; redesigned the DB layer for **50MB+** contest test cases.
- Built AI voice interviews (**LiveKit**): recruiter, behavioral, and coding agents run from versioned, admin-authored definitions.
- Moved CI to **Vercel** when Actions quota ran out: one fail-fast pipeline (format, typecheck, test, build), no prod secrets.
- Tracked down live-contest outages with **Claude Code**, **MCP servers**, **PostHog**, and log analysis, then shipped the fixes.
- Built a real-time collaborative dev space using **WebSockets** and **Yjs CRDTs** over a Monaco live-syncing editor.

## OPEN SOURCE CONTRIBUTIONS

- **GitLab Go SDK (client-go)**: typed group integrations – Slack, Jira, MS Teams, Mattermost, Harbor, +3 | [!2692](#), [!2691](#), [!2675](#), [!2670](#).
- **GitLab Terraform provider**: new Harbor & Mattermost integration resources + group SAML attribute | [!2881](#), [!2909](#), [!2852](#).
- **GitLab monorepo**: `Json.safe_parse` hardening and Vue lint compliance | [!221163](#), [!221158](#).
- **LLVM / Clang**: `constexpr X86 cvtpd2ps`, `AArch64/ARM CLS` → `ISD`: `:CTLS`, `LLDB SWIG` tests | [#169980](#), [#178885](#), [#178430](#), [#173473](#).
- **Apple Swift NIO**: idempotent `NIOFileSystem` directory creation | [#3410](#).

## PROJECTS

Autonomous AI Tutor & Curriculum Engine, Graduation Project github.com/sigma-loop  
• Led backend & DevOps for an 8-person team building an on-demand AI course, lesson, and challenge generator.  
• Graded code in a Judge0 sandbox and math proofs via a confidence-scored LLM; owned all CI/CD and AWS deployments.

Competitive-Programming Judge — AWS Reference Architecture github.com/HamzaHassanain/CP-Judge-AWS  
• Designed an AWS reference architecture: Judge0 `isolate` on ECS-on-EC2, Fargate orchestrator, Spot ASG on queue depth.  
• Automated post-round cheating detection with Step Functions, AWS Batch, Lambda/Bedrock, and Athena.

Polyman CLI npmjs.com/package/polyman-cli | github.com/HamzaHassanain/polyman  
• Built a TypeScript CLI automating the Codeforces Polygon workflow: test generation, validation, and direct API sync.

Systems in C++ — HALlocator & cppress HALlocator | cppress  
• Built a Red-Black-Tree best-fit memory allocator ( $O(\log n)$ ) and an epoll-driven C++ web framework, both from scratch.

## TECHNICAL WRITING

- Authored [”What Every Programmer Should Know About Memory”](#), a series on CPU caches, virtual memory, and optimization.
- Published [”I Built a Backend Web Framework from Scratch in C++”](#), covering non-blocking I/O, HTTP parsing, and thread pools.

## ACHIEVEMENTS

ICPC ECPC & ACPC — Competitive Programming icpc.global/YC7Z02WQCPIB  
*17th of 500 teams at ECPC (2024); two-time ACPC regional qualifier (2024, 2025), top 250 in the Africa & Arab region.*

## SKILLS

|                      |                                                                                                          |
|----------------------|----------------------------------------------------------------------------------------------------------|
| Cloud & DevOps       | AWS, Docker, Nginx, Redis, Terraform, CI/CD (GitHub Actions), DigitalOcean, Linux                        |
| Backend & Full-Stack | TypeScript, Node.js, Express.js, Next.js, React, MongoDB, PostgreSQL, REST APIs, WebSockets, Yjs (CRDTs) |
| Languages & Systems  | C++, Go, Rust, Python, SQL; System Design, Distributed Systems, Message Queues                           |